

Pranav Parab

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EXPERIENCES

Data Engineer

(April 2024 - Present)

Celebal Technologies

- Built real-time CDC ingestion pipelines in **Databricks** using **Auto Loader** and **Spark Structured Streaming**, delivering **10M+ records/day** into **Delta Lake** with **sub-minute latency** for operational dashboards.
- Developed LLM-powered automation tool using Python, **OpenAI GPT-4**, and **Claude** to convert Scala to PySpark, **reducing migration effort by 80%**.
- Led 4-member team migrating data to **Databricks Unity Catalog** using **Terraform**, SQL, PySpark, enabling governance and achieving **30% infrastructure reduction**.
- Architected **medallion lakehouse** (Bronze/Silver/Gold) for healthcare client with Databricks Inc., implementing **Delta Live Tables (DLT)** pipelines with declarative expectations to enforce data quality constraints and **automate lineage tracking, reducing processing time by 40%**.
- Led 5-member team building **AI-powered analytics migration tool** using **Claude**. Developed **Evolutionary Mapper algorithm**, saving **~50 hours per dashboard**.

Research Data Engineer

(September 2021 - April 2024)

SIRJ Lab, University of Maryland

- Built data pipelines on **AWS** (Lambda, RDS, Redshift) and **GCP** (Cloud Functions, BigQuery) using serverless architecture, **reducing infrastructure costs by 99%**.
- Orchestrated end-to-end workflows with **Apache Airflow**, automating ingestion, transformation, and loading into **Snowflake** with alerting and retry logic.
- Built interactive dashboards in **Power BI**, **Tableau**, and **Looker Studio**, **improving stakeholder analysis efficiency by 40%**.

ETL Engineer

(July 2018 - May 2021)

Larsen & Toubro Infotech Limited

- Engineered multi-tier **ETL pipelines** for claims data processing using **Ab Initio**, **Databricks**, **SSIS**, and Python, integrating PostgreSQL, Teradata, and SQL Server for enterprise data warehousing.
- Developed **data quality framework** with stored procedures and validation functions, **reducing data discrepancies by 20%**.
- Collaborated in Agile sprints with cross-functional teams, contributing to on-time delivery.

PROJECTS

- **Stock Market Real-Time Data Analysis tool:** Developed a data pipeline integrating Python, Kafka, Amazon S3, EC2, AWS Glue, Athena, and Tableau for real-time stock market data analysis and visualization.
- **AI-Powered Nonprofit Content Agent:** Built full-stack **AI platform** using Python, JavaScript, **Google Gemini**, and **Claude**. Designed **model routing architecture** with prefetch algorithm, **reducing latency from 25s to 100ms**.

SKILLS

Data Engineering: Apache Spark, Kafka, Airflow, Hive, Hadoop, Python, SQL, PySpark, Scala, Delta Lake, dbt, Structured Streaming, Data Modeling, Star Schema, Data Observability

Databases: Snowflake, Databricks, BigQuery, Redshift, PostgreSQL, MySQL, SQL Server, MongoDB, Teradata

Cloud: AWS (Lambda, Glue, S3, EC2, RDS, Athena, EMR), GCP (BigQuery, Dataflow), Azure (Data Lake Gen2, Synapse)

AI/ML: OpenAI GPT-4, Claude, Google Gemini, Prompt Engineering, LLM Automation, Model Routing

Tools: Git, Docker, Kubernetes, Terraform, CI/CD, SSIS, Ab Initio, Tableau, Power BI, Looker Studio, Jira

EDUCATION

M.S. in Computer Software Engineering – University of Maryland, College Park, USA

B.S. in Computer Science – University of Mumbai